



2 Real Analysis 2023

Tutorial 11

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Images and preimages

Problem 44 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \sin x$. Determine the following images and preimages:

$$f\left(\left[0, \frac{\pi}{2}\right]\right), f\left(\left\{\frac{\pi}{4}\right\}\right), f(\mathbb{R}), f^{-1}(\{0\}), f^{-1}([0, 1]), f^{-1}((2, 3)), f^{-1}(\mathbb{R}).$$

Solution.

$$\text{We have } f\left(\left[0, \frac{\pi}{2}\right]\right) = [0, 1], f\left(\left\{\frac{\pi}{4}\right\}\right) = \left\{\frac{1}{\sqrt{2}}\right\}, f(\mathbb{R}) = [-1, 1],$$

$$f^{-1}(\{0\}) = \{\pi k : k \in \mathbb{Z}\}, f^{-1}([0, 1]) = \bigcup_{k \in \mathbb{Z}} [2\pi k, 2\pi k + \pi], f^{-1}((2, 3)) = \emptyset, f^{-1}(\mathbb{R}) = \mathbb{R}.$$

Problem 45 Let X, Y be nonempty sets and $f : X \rightarrow Y$. Prove the following statements.

- (a) $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$, $A_1, A_2 \subseteq X$.
- (b) $f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$, $B_1, B_2 \subseteq Y$.
- (c) $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$, $A_1, A_2 \subseteq X$.
- (d) $f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$, $B_1, B_2 \subseteq Y$.
- (e) $A \subseteq f^{-1}(f(A))$, $A \subseteq X$.
- (f) $f(f^{-1}(B)) \subseteq B$, $B \subseteq Y$.

Use the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, to show that the sign $\subseteq$ in parts (c), (e) and (f) cannot be replaced by $=$.

Solution.

- (a) If $y \in f(A_1 \cup A_2)$, then $y = f(x)$ with some $x \in A_1 \cup A_2$. If $x \in A_1$, then $y \in f(A_1)$, and if $x \in A_2$, then $y \in f(A_2)$. Thus, $y \in f(A_1) \cup f(A_2)$. This shows that $f(A_1 \cup A_2) \subseteq f(A_1) \cup f(A_2)$.

Conversely, let $y \in f(A_1) \cup f(A_2)$. Then $y \in f(A_1)$ or $y \in f(A_2)$. In the first case there is $x \in A_1$ such that $y = f(x)$, and in the second case there is $x \in A_2$

such that $y = f(x)$. We see that $y = f(x)$ with $x \in A_1 \cup A_2$. This proves that $f(A_1 \cup A_2) \supseteq f(A_1) \cup f(A_2)$.

The two inclusions together give $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$.

- (b) If $x \in f^{-1}(B_1 \cup B_2)$, then $f(x) \in B_1 \cup B_2$. If $f(x) \in B_1$, then $x \in f^{-1}(B_1)$, and if $f(x) \in B_2$, then $x \in f^{-1}(B_2)$. Thus, $x \in f^{-1}(B_1) \cup f^{-1}(B_2)$. This proves that $f^{-1}(B_1 \cup B_2) \subseteq f^{-1}(B_1) \cup f^{-1}(B_2)$.

Conversely, let $x \in f^{-1}(B_1) \cup f^{-1}(B_2)$. If $x \in f^{-1}(B_1)$, then $f(x) \in B_1$, and if $x \in f^{-1}(B_2)$, then $f(x) \in B_2$. Thus, $f(x) \in B_1 \cup B_2$. This implies that $x \in f^{-1}(B_1 \cup B_2)$. Therefore, $f^{-1}(B_1 \cup B_2) \supseteq f^{-1}(B_1) \cup f^{-1}(B_2)$.

The two inclusions together give $f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$.

- (c) If $y \in f(A_1 \cap A_2)$, then $y = f(x)$ with some $x \in A_1 \cap A_2$. Thus, $y \in f(A_1)$ and $y \in f(A_2)$. This implies $y \in f(A_1) \cap f(A_2)$. This proves the inclusion $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$.

To show that $\subseteq$ cannot be replaced by $=$, we consider $f(x) = x^2$, $A_1 = [-1, 0]$ and $A_2 = [0, 1]$. Then $f(A_1) = f(A_2) = [0, 1]$ and $f(A_1) \cap f(A_2) = [0, 1]$. But $A_1 \cap A_2 = \{0\}$ and $f(A_1 \cap A_2) = \{0\}$.

- (d) If $x \in f^{-1}(B_1 \cap B_2)$, then $f(x) \in B_1 \cap B_2$, i.e. $f(x) \in B_1$ and $f(x) \in B_2$. This implies $x \in f^{-1}(B_1)$ and $x \in f^{-1}(B_2)$. Therefore, $x \in f^{-1}(B_1) \cap f^{-1}(B_2)$. This proves the inclusion $f^{-1}(B_1 \cap B_2) \subseteq f^{-1}(B_1) \cap f^{-1}(B_2)$.

Conversely, if $x \in f^{-1}(B_1) \cap f^{-1}(B_2)$, then $x \in f^{-1}(B_1)$ and $x \in f^{-1}(B_2)$. Hence, $f(x) \in B_1$ and $f(x) \in B_2$. Consequently, $f(x) \in B_1 \cap B_2$, which means that $x \in f^{-1}(B_1 \cap B_2)$. This shows the inclusion $f^{-1}(B_1 \cap B_2) \supseteq f^{-1}(B_1) \cap f^{-1}(B_2)$.

The two inclusions together give $f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$.

- (e) If $x \in A$, then $f(x) \in f(A)$, and therefore $x \in f^{-1}(f(A))$. It follows that $A \subseteq f^{-1}(f(A))$.

To see that $\subseteq$ cannot be replaced by $=$, we consider $f(x) = x^2$ and $A = [0, 1]$. Then $f(A) = [0, 1]$ and $f^{-1}(f(A)) = [-1, 1]$.

- (f) If $y \in f(f^{-1}(B))$, then there is $x \in f^{-1}(B)$ such that $y = f(x)$. But $x \in f^{-1}(B)$ means that $f(x) \in B$. We see that $y \in B$. This proves the inclusion $f(f^{-1}(B)) \subseteq B$.

To see that $\subseteq$ cannot be replaced by $=$, we consider $f(x) = x^2$ and $B = [-1, 1]$. Then $f^{-1}(B) = [0, 1]$ and $f(f^{-1}(B)) = [0, 1]$.

Continuous functions

Problem 46 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at point $x_0 \in \mathbb{R}$. Show that if $f(x_0) > 0$, then there exists $\delta > 0$ such that $f(x) > 0$ for all $x \in (x_0 - \delta, x_0 + \delta)$.

Solution.

By the definition of continuity with $\varepsilon = \frac{1}{2}f(x_0) > 0$, there exists $\delta > 0$ such that

$$|f(x) - f(x_0)| < \frac{1}{2}f(x_0) \quad \text{for all } x \in (x_0 - \delta, x_0 + \delta).$$

For $x \in (x_0 - \delta, x_0 + \delta)$ we have

$$f(x) > f(x_0) - \frac{1}{2}f(x_0) = \frac{1}{2}f(x_0) > 0.$$

Problem 47 Suppose that two functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are both continuous at point $x_0 \in \mathbb{R}$. Prove by definition that their sum $f + g$ is continuous at x_0 .

Solution.

Given $\varepsilon > 0$, there exist $\delta_1, \delta_2 > 0$ such that

$$\begin{aligned} |f(x) - f(x_0)| &< \frac{\varepsilon}{2} \quad \text{for all } x \in (x_0 - \delta_1, x_0 + \delta_1), \\ |g(x) - g(x_0)| &< \frac{\varepsilon}{2} \quad \text{for all } x \in (x_0 - \delta_2, x_0 + \delta_2). \end{aligned}$$

Put $\delta = \min\{\delta_1, \delta_2\} > 0$. Then for all $x \in (x_0 - \delta, x_0 + \delta)$ we have

$$\begin{aligned} |(f + g)(x) - (f + g)(x_0)| &= |f(x) + g(x) - f(x_0) - g(x_0)| \\ &\leq |f(x) - f(x_0)| + |g(x) - g(x_0)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus, by definition, $f + g$ is continuous at x_0 .

Problem 48 Give an example of two functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are both discontinuous at a certain point $x_0 \in \mathbb{R}$, but both the sum $f + g$ and the product fg are continuous at x_0 .

Solution.

Consider, for example,

$$f(x) = \begin{cases} 1, & x \geq 0, \\ -1, & x < 0, \end{cases} \quad \text{and} \quad g(x) = -f(x) = \begin{cases} -1, & x \geq 0, \\ 1, & x < 0. \end{cases}$$

Both functions f and g are discontinuous at 0, but $f + g \equiv 0$ and $fg \equiv -1$ are continuous everywhere.

Problem 49 Consider the function

$$f(x) = \begin{cases} 2x, & x \in \mathbb{Q}, \\ x + 3, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Determine all points $x \in \mathbb{R}$ where f is continuous.

Solution.

For any $x \in \mathbb{R}$ there is a sequence $(r_n)_{n=1}^\infty$ in $\mathbb{Q}$ and a sequence $(t_n)_{n=1}^\infty$ in $\mathbb{R} \setminus \mathbb{Q}$ such that $\lim_{n \rightarrow \infty} r_n = \lim_{n \rightarrow \infty} t_n = x$. If f is continuous at x , we must have

$$\lim_{n \rightarrow \infty} f(r_n) = \lim_{n \rightarrow \infty} f(t_n).$$

We have $\lim_{n \rightarrow \infty} f(r_n) = 2x$ and $\lim_{n \rightarrow \infty} f(t_n) = x + 3$. Thus, f is continuous at x if and only if $2x = x + 3$. We conclude that f is continuous only at $x = 3$.

Uniform continuity

Problem 50 Consider the function $f(x) = \frac{1}{1-x}$, $x \in \mathbb{R} \setminus \{1\}$.

- (a) Construct two sequences $(x_n)_{n=1}^\infty$, $(y_n)_{n=1}^\infty$ in $(0, 1)$ such that $\lim_{n \rightarrow \infty} (x_n - y_n) = 0$, but $|f(x_n) - f(y_n)| = 1$ for all $n \in \mathbb{N}$. Deduce that f is not uniformly continuous on $(0, 1)$.
- (b) Show that f is uniformly continuous on $[0, \frac{1}{2}]$.

Solution.

- (a) Take $x_n = 1 - \frac{1}{n+1}$, $y_n = 1 - \frac{1}{n+2}$, $n \in \mathbb{N}$. Then

$$x_n - y_n = \left(1 - \frac{1}{n+1}\right) - \left(1 - \frac{1}{n+2}\right) = \frac{1}{n+2} - \frac{1}{n+1} \rightarrow 0, \quad n \rightarrow \infty,$$

but $f(x_n) = n+1$ and $f(y_n) = n+2$, so that $|f(x_n) - f(y_n)| = 1$ for all $n \in \mathbb{N}$.

It follows by Proposition 4.28 that f is not uniformly continuous on $(0, 1)$.

- (b) Let $x, t \in [0, \frac{1}{2}]$. We have

$$|f(x) - f(t)| = \left| \frac{1}{1-x} - \frac{1}{1-t} \right| = \frac{|x-t|}{|1-x||1-t|}.$$

For $x, t \in [0, \frac{1}{2}]$ we have $|1-x| \geq \frac{1}{2}$, $|1-t| \geq \frac{1}{2}$, and thus $\frac{1}{|1-x||1-t|} \leq 4$. We conclude that

$$|f(x) - f(t)| \leq 4|x-t|.$$

Given $\varepsilon > 0$, take $\delta = \frac{\varepsilon}{4} > 0$. Then for any $x, t \in [0, \frac{1}{2}]$

$$|x-t| < \delta \implies |f(x) - f(t)| \leq 4 \cdot \frac{\varepsilon}{4} = \varepsilon.$$

This implies that f is uniformly continuous on $[0, \frac{1}{2}]$.